11. If f, g, h are real valued functions defined on $\mathbb{R}$, then prove that $(f+g)\circ h = f\circ h + g\circ h$. What can you say about $f\circ(g+h)$? Justify your answer.
12. If$f:\mathbb{R}\to\mathbb{R}$ is defined by $f(x)=3x-5$, prove that f is a bijection and find its inverse.
13. The weight of the muscles of a man is a function of his body weight x and can be expressed as $W(x)=0.35x$. Determine the domain of this function.
14. The distance of an object falling is a function of time t and can be expressed as $s(t)=-16t^2$. Graph the function and determine if it is one-to-one.
15. The total cost of airfare on a given route is comprised of the base cost C and the fuel surcharge S in rupee. Both C and S are functions of the mileage m; $C(m)=0.4m+50$ and $S(m)=0.03m$. Determine a function for the total cost of a ticket in terms of the mileage and find the airfare for flying 1600 miles.
16. A salesperson whose annual earnings can be represented by the function $A(x)=30,000+0.04x$, where x is the rupee value of the merchandise he sells. His son is also in sales and his earnings are represented by the function $S(x)=25,000+0.05x$. Find $(A+S)(x)$ and determine the total family income if they each sell Rupees $1,50,00,000$ worth of merchandise.
17. The function for exchanging American dollars for Singapore Dollar on a given day is $f(x)=1.23x$, where x represents the number of American dollars. On the same day the function for exchanging Singapore Dollar to Indian Rupee is $g(y)=50.50y$, where y represents the number of Singapore dollars. Write a function which will give the exchange rate of American dollars in terms of Indian rupee.
18. The owner of a small restaurant can prepare a particular meal at a cost of Rupees 100. He estimates that if the menu price of the meal is x rupees, then the number of customers who will order that meal at that price in an evening is given by the function $D(x)=200-x$. Express his day revenue, total cost and profit on this meal as functions of x.
19. The formula for converting from Fahrenheit to Celsius temperatures is $y=\dfrac{5x}{9}-\dfrac{160}{9}$. Find the inverse of this function and determine whether the inverse is also a function.
20. A simple cipher takes a number and codes it, using the function $f(x)=3x-4$. Find the inverse of this function, determine whether the inverse is also a function and verify the symmetrical property about the line $y=x$ (by drawing the lines).

1.7 Graphing Functions using Transformations

"*A picture is worth a thousand words*" is a well known proverb. To know about a function well, its graph will help us more than its analytical expression. To draw graphs quickly without plotting many points is an invaluable skill. Familiarity with shapes of some basic functions will help to graph other complicated functions. Understanding and usage of symmetry and transformations will then enable to strengthen graphing abilities. This section is not simply a data base of graphs, we learn some methods to graph certain functions.

Suppose that we want to draw or sketch the curve of the function $y=2\sin(x-1)+3$. At the very first sight it looks that it is very difficult to draw the curve representing this function. But it will be very easy to draw after understanding the content of this section.

If we know a half of a graph is the mirror image of the other half with respective to a line, or a graph can be obtained just by moving a known graph in some direction, then we can draw the new one using the known one. Moreover if we know that a graph can be obtained by enlarging or shrinking a known one, then also we can draw the new one using the known ones.

The following type of transformations play very important roles in graphing.

(i) Reflection
(ii) Translation
(iii) Dilations.